

190. Reverse Bits

Easy

Topics

Companies

Reverse bits of a given 32 bits signed integer.

Example 1:

Input: n = 43261596

Output: 964176192

Explanation:

Integer	Binary
43261596	00000010100101000001111010011100
964176192	00111001011110000010100101000000

Example 2:

Input: $n = 2147483644$

Output: 1073741822

Explanation:

Integer	Binary
2147483644	0111111111111111111111111111100
1073741822	0011111111111111111111111111110

Constraints:

- $0 \leq n \leq 2^{31} - 2$
- n is even.

Follow up: If this function is called many times, how would you optimize it?

Python:

class Solution:

def reverseBits(self, n: int) -> int:

result = 0

for _ in range(32):

Shift result left and add the last bit of n

result = (result << 1) | (n & 1)

Shift n right to process the next bit

n >>= 1

return result

JavaScript:

```
/**
 * @param {number} n
 * @return {number}
 */
var reverseBits = function(n) {
    let result = 0;

    for (let i = 0; i < 32; i++) {
        result <<= 1;    // Shift result left to make space
        result |= (n & 1); // Add the last bit of n
        n >>= 1;         // Drop the last bit of n
    }

    // >>> 0 ensures unsigned 32-bit integer
    return result >>> 0;
};
```

Java:

```
class Solution {
    public int reverseBits(int n) {
        int result = 0;

        for (int i = 0; i < 32; i++) {
            // Shift result left to make space
            result <<= 1;

            // Add the last bit of n
            result |= (n & 1);

            // Shift n right (unsigned shift to avoid sign extension)
            n >>= 1;
        }

        return result;
    }
}
```